Project Report

On

Design and Implementation of a 4-Bit Comparator Circuit

Course No: ECE 2104

Course Name: Digital Electronics and Logic Circuits Laboratory

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1. Detailed specification of the project

1.1 Abstract

This project focuses on the design and implementation of a 4-bit comparator circuit using digital logic gates and basic combinational circuit techniques. The purpose of the project is to compare two 4-bit binary numbers and determine their relationship. The comparator circuit produces three different outputs to indicate whether the first binary number is greater than the second number (A > B), equal to the second number (A = B), or less than the second number (A < B).

In this project, LEDs are used as output indicators. Three separate LEDs are connected to the output lines of the comparator circuit, where each LED represents one comparison condition. When a particular condition becomes true, the corresponding LED turns ON, providing a clear visual indication of the comparison result. This method makes the circuit simple, cost-effective, and easy to analyze during practical implementation.

The circuit is designed based on Boolean algebra and combinational logic concepts. Different logic gates such as AND, OR, NOT, and XNOR gates are used to develop the complete comparator system. DIP switches are used to provide binary inputs for the two 4-bit numbers, while TTL logic ICs are used for hardware implementation. The project was first analyzed theoretically and then implemented practically using electronic components on a breadboard or simulation software.

The main goal of this project is to develop a clear understanding of digital comparison operations, logic gate implementation, truth table analysis, and practical digital circuit design. During testing, different combinations of binary inputs were applied to the circuit, and the corresponding LEDs successfully indicated the correct comparison results. The experimental outputs matched the expected theoretical results, proving that the designed 4-bit comparator circuit operates accurately and efficiently according to the required specifications.

1.2 Methodology

The methodology of this project is based on the design, implementation, and testing of a 4-bit comparator circuit using digital logic gates. The complete work was carried out through several systematic steps to ensure proper operation of the circuit.

1. **Problem Identification and Objective Selection**

At first, the objective of the project was determined. The main goal was to design a digital circuit capable of comparing two 4-bit binary numbers and generating three different outputs indicating:

- A > B (Greater Than)
- A = B (Equal To)
- A < B (Less Than)

The output indications were planned using three LEDs for simple and clear visualization.

2. **Theoretical Analysis**

The working principle of binary comparators and combinational logic circuits was studied. Truth tables and Boolean expressions were analyzed to understand how the comparison process works between two binary numbers. The logic conditions required for greater-than, equal-to, and less-than operations were derived mathematically.

3. **Circuit Design**

Based on the Boolean expressions, the comparator circuit was designed using digital logic gates such as:

- AND gates
- OR gates
- NOT gates
- XNOR gates

Two sets of 4-bit binary inputs were considered as input variables $A_3A_2A_1A_0$ and $B_3B_2B_1B_0$. The outputs of the logic gates were connected to generate the final comparison outputs.

4. **Component Selection**

Necessary electronic components were selected for hardware implementation. The major components used in the project include:

- Logic gate ICs
- DIP switches for binary inputs
- LEDs for output indication
- Current limiting resistors
- Breadboard and connecting wires
- Power supply

5. **Circuit Implementation**

The designed circuit was implemented practically on a breadboard. DIP switches were connected to provide input combinations, and LEDs were connected to the output terminals through resistors to display the comparison results. Proper power supply and grounding connections were ensured during implementation.

6. **Testing and Verification**

Different combinations of 4-bit binary inputs were applied to the circuit. The outputs were observed carefully through the LEDs to verify whether the circuit correctly identified greater-than, equal-to, and less-than conditions. The practical outputs were then compared with the theoretical truth table results.

2. Design of the project

2.1 Block Diagram Overview

The system is built around three functional blocks arranged in cascade

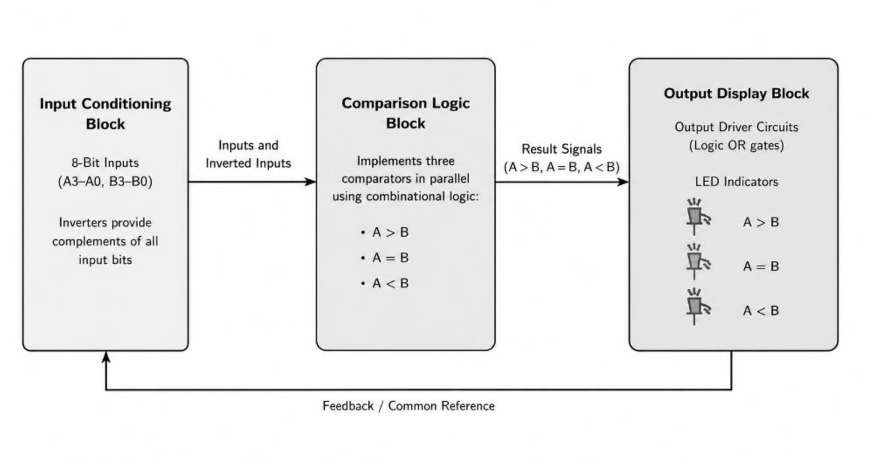

Fig 1: Block diagram of a 4 bit comparator circuit

2.2 Circuit Schematic

The complete circuit schematic designed in Proteus is shown below

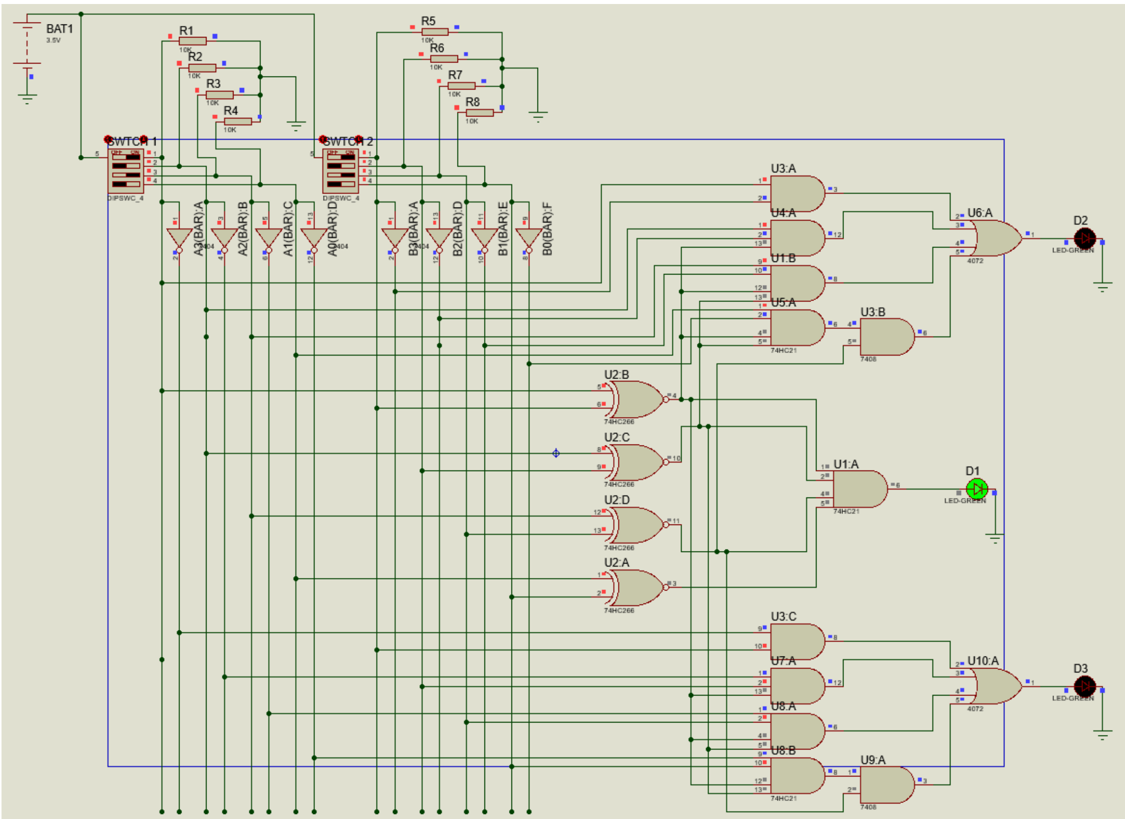

Fig 2: Circuit schematic of a 4 bit comparator circuit

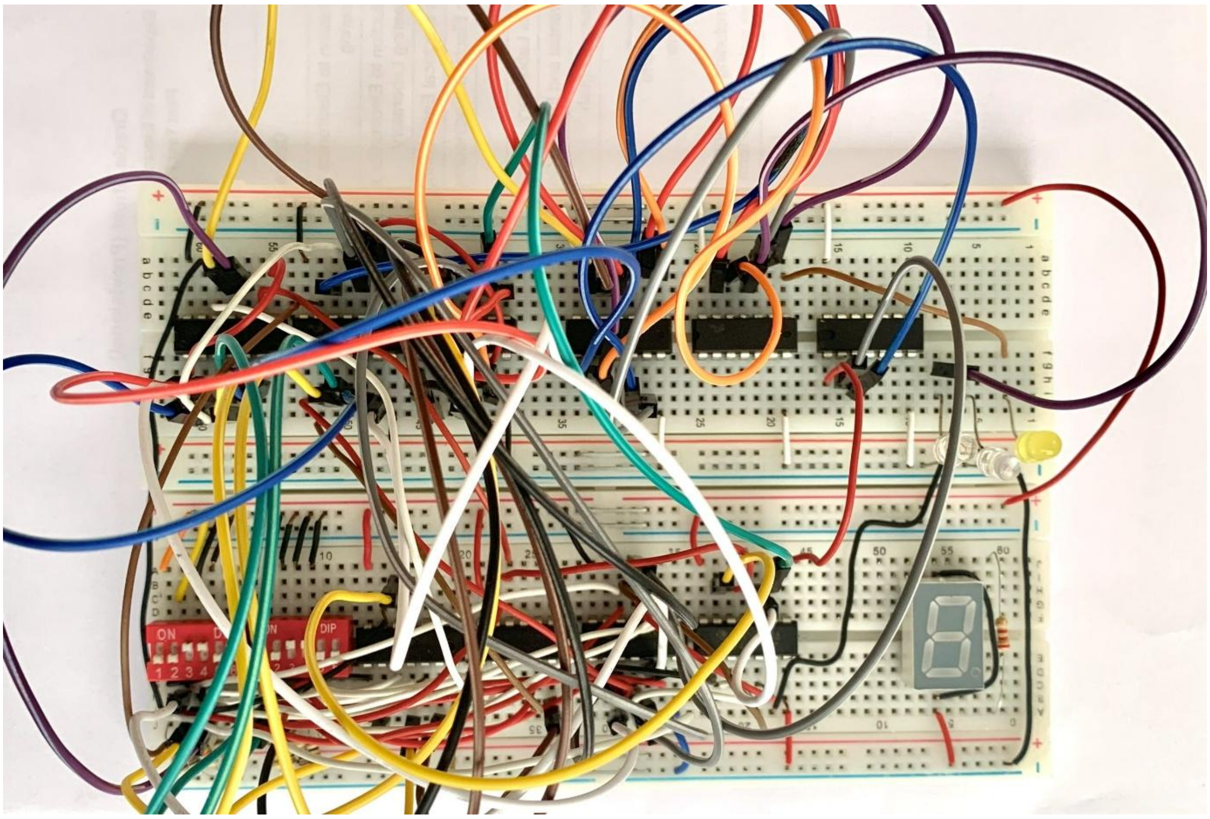

Fig 3: The Final setup[Breadboard Implimentation]

3. Component List and Specifications

Table 1: Below lists all components used in the physical implementation of the circuit:

Sl.	Item	Justification	Approx. Price (BDT)
1	IC 7404 (Hex NOT Gate) ×2	Used for input inversion in comparator logic	80
2	IC 7408 (Quad 2-input AND Gate) ×2	Used for implementing AND logic operations	80
3	IC 7411 (Triple 3-input AND Gate) ×2	Used for multi-input comparison logic	100
4	IC 7421 (Dual 4-input AND Gate) ×3	Used for higher-order AND operations	180
5	IC 4072 (Dual 4-input OR Gate) ×2	Used for OR logic in output generation	120

6	IC 7486 (Quad XOR Gate) ×2	Used for equality comparison operations	100
7	DIP Switch (4-bit) ×2	Used for providing binary input values	120
8	LED ×3	Used for displaying A>B, A=B, and A<B conditions	30
9	10kΩ Resistor ×8	Used as pull-up/pull-down resistors	40
10	220Ω Resistor ×3	Used for current limiting of LEDs	15
11	Breadboard	Used for circuit prototyping and implementation	180
12	Connecting Wires	Used for electrical connections	100
13	DC Power Supply (+5V)	Used for powering TTL logic ICs	250
14	Battery (3.5V/5V source)	Portable power source for testing	80
15	7-Segment Display	Used for displaying output result	40
Total (BDT)			**1505**
In words: One thousand five hundred five taka only			

4. Result Analysis

The designed 4-bit comparator circuit was implemented and tested successfully using different combinations of binary inputs. Two separate 4-bit binary numbers were provided through DIP switches, and the circuit compared these inputs continuously using combinational logic operations. The comparison results were displayed through three LEDs representing the conditions A > B, A = B, and A < B.

During the experiment, several input combinations were applied to verify the accuracy and reliability of the circuit. When the value of input A was greater than input B, the LED corresponding to the “A > B” condition turned ON correctly. Similarly, when both binary numbers were equal, only the “A = B” LED glowed, indicating that the comparator detected equality successfully. In cases where input A was smaller than input B, the “A < B” LED became active as expected.

The observed outputs matched the theoretical truth table and Boolean expressions used during circuit design. At any particular time, only one LED remained active, which confirmed that the comparator logic worked properly without output conflict. The circuit also showed quick response to input changes and maintained stable operation throughout the testing process.

The hardware implementation demonstrated that digital logic gates can effectively perform binary comparison operations with high accuracy. The use of LEDs as output indicators made the comparison results simple and easy to understand visually. No major functional errors were observed during testing, although minor connection issues and troubleshooting were required during circuit setup.

Overall, the experimental results proved that the designed 4-bit comparator circuit fulfilled the project objectives successfully. The project provided practical knowledge about combinational logic design, logic gate implementation, Boolean algebra, and digital hardware testing techniques.

5. Detailed description and technical issues encountered

During the design and implementation of the 4-bit comparator circuit, several practical and technical problems were encountered. These issues mainly occurred during circuit design, hardware connection, component arrangement, and output verification stages. Different troubleshooting techniques were applied to solve these problems and ensure proper operation of the circuit.

1. Complexity of Logic Design

One of the main challenges was designing the comparator logic correctly using multiple logic gates. Since the circuit compares two 4-bit binary numbers, the Boolean expressions for greater-than, equal-to, and less-than conditions became lengthy and complex. Extra attention was required while simplifying and implementing the logic equations to avoid incorrect outputs.

2. Incorrect Wiring Connections

During hardware implementation, several wiring mistakes occurred on the breadboard. Incorrect placement of IC pins, loose jumper wire connections, and improper grounding initially caused unexpected outputs. These problems made troubleshooting difficult because even a small connection error affected the entire circuit operation. The issue was solved by carefully checking each connection according to the circuit schematic.

3. LED Output Problems

At first, the LEDs did not operate correctly due to improper resistor connections and polarity issues. In some cases, more than one LED turned ON simultaneously because of incorrect output connections. Proper current limiting resistors and correct LED polarity were later ensured to obtain accurate output indications.

4. IC Identification and Pin Configuration Issues

Another challenge was identifying the correct pin numbers and functions of different logic gate ICs. Confusion between input pins, output pins, VCC, and GND connections sometimes caused the ICs not to function properly. The datasheets of the ICs were studied carefully to solve this problem.

5. Power Supply Stability

The comparator circuit required a stable +5V power supply for proper TTL logic operation. At certain stages, unstable voltage supply and loose power connections caused inconsistent outputs and LED flickering. This issue was solved by ensuring proper power distribution and secure connections on the breadboard.

6. Testing and Verification Difficulties

Testing the circuit with multiple input combinations was time-consuming because every possible condition needed to be verified manually. Sometimes the observed outputs did not match the expected theoretical results due to wiring errors or logic gate connection mistakes. Rechecking the truth table and debugging the circuit step by step helped resolve these issues.

7. Breadboard Limitations

As the circuit used several logic gates and connecting wires, the breadboard became crowded, making the circuit difficult to organize. This increased the possibility of accidental disconnections and short circuits. Proper wire management and careful component placement were necessary to maintain a stable setup.

Despite these technical difficulties, all problems were gradually identified and solved through repeated testing, troubleshooting, and verification. Overcoming these issues helped in gaining practical knowledge about digital electronics, logic circuit implementation, hardware debugging, and problem-solving techniques.

5. Conclusions

5.1 Whether the project works according to provided specifications

The implemented 4-bit comparator circuit operated successfully according to the provided project specifications. The main objective of the project was to compare two 4-bit binary numbers and generate three different output conditions indicating whether the first number is greater than, equal to, or less than the second number. The circuit achieved all of these objectives accurately.

During testing, different combinations of binary inputs were applied through DIP switches, and the outputs were observed using three LEDs. The LEDs correctly indicated the comparison results for all tested conditions:

- The “first” LED turned ON when the value of A was greater than B.
- The “second” LED turned ON when both binary numbers were equal.
- The “third” LED turned ON when the value of A was smaller than B.

The outputs matched the expected truth table and theoretical calculations. The circuit also responded properly to changes in input combinations without noticeable delay or instability. At any moment, only one output LED remained active, confirming the correct implementation of comparator logic.

The hardware implementation using logic gate ICs, DIP switches, resistors, and LEDs functioned reliably under the required operating conditions. Therefore, it can be concluded that the project fulfilled its intended purpose and worked according to the provided specifications successfully.

5.2 Exceptions

Although the 4-bit comparator circuit operated successfully according to the project requirements, a few limitations and exceptional conditions were observed during implementation and testing.

The circuit was designed mainly for educational and demonstration purposes, so it was implemented using basic logic gates and discrete components. As a result, the hardware arrangement became comparatively complex due to the large number of ICs and wiring connections. In larger digital systems, dedicated comparator ICs could reduce circuit complexity and improve efficiency.

Another limitation was that the project used LEDs only for output indication. Therefore, the circuit could display only one comparison condition at a time and did not provide any numerical or advanced visual output representation. In addition, the LEDs depended on proper voltage and current conditions; otherwise, brightness variations or incorrect indications could occur.

Minor errors also appeared during practical testing because of loose breadboard connections, wiring mistakes, and power supply instability. These issues did not affect the final functionality permanently but required repeated troubleshooting and correction during implementation.

The circuit performance was tested under standard laboratory conditions with a regulated +5V power supply. The behavior of the circuit under different environmental conditions, noise, or higher operating frequencies was not analyzed in this project.

Despite these exceptions and limitations, the comparator circuit performed its intended comparison operation correctly and produced accurate outputs for the tested input combinations.

5.3 Ways to improve the project

The implemented 4-bit comparator circuit performed successfully; however, several improvements can be made to increase its efficiency, functionality, and practical usability in future development.

One possible improvement is to reduce the circuit complexity by using dedicated comparator ICs such as the 7485 IC instead of building the entire system with multiple individual logic gates. This would decrease the number of components, simplify wiring connections, and improve circuit reliability.

The project can also be improved by adding a seven-segment display or LCD display to show the comparison result more clearly and professionally instead of using only LEDs. This would make the output easier to understand, especially for larger or more advanced digital systems.

Another improvement would be implementing the circuit using simulation and PCB design software to create a more compact and permanent hardware setup. A printed circuit board (PCB) would reduce loose connections, minimize wiring errors, and improve overall circuit stability.

The comparator can further be extended from a 4-bit system to an 8-bit or higher-bit comparator for handling larger binary numbers. This enhancement would increase the practical applications of the project in modern digital systems and processors.

Additional features such as automatic input generation, microcontroller integration, or programmable logic implementation using FPGA or CPLD technology can also be included to make the project more advanced and efficient.

Due to time limitations and hardware constraints, the planned integration of a 7-segment display could not be completed during the implementation phase of the project. In future development, a seven-segment display can be added to provide a clearer and more professional visualization of the comparison results. Additionally, the comparison process can be implemented using conditional programming techniques such as if–else statements through microcontroller integration, which would make the system more flexible, efficient, and easier to expand for advanced applications.

Finally, improving power supply regulation and using better-quality components could increase the accuracy and long-term stability of the circuit. These modifications would help transform the project from a basic educational model into a more practical and industry-oriented digital system.

References

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Github Repository: https://github.com/saiful-islam-sazzat/4-bit-Comparator-DELC-2104-project-